3D Printing of Polymer Nanocomposites via Stereolithography

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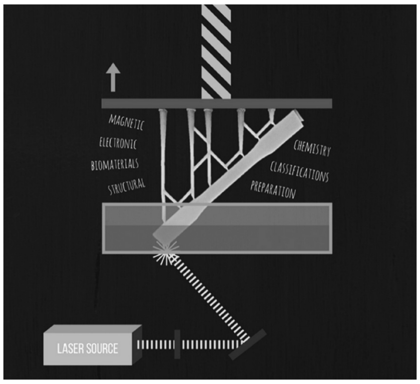
Additive manufacturing (AM) is still underutilized as an industrial process, but is quickly gaining momentum with the development of innovative techniques and materials for various applications. In particular, stereolithography (SLA) is now shifting from rapid prototyping to rapid manufacturing, but is facing challenges in parts performance and printing speed, among others. This review discusses the application of SLA for polymer nanocomposites fabrication to show the technology's potential in increasing the applicability of current SLA-printed parts. Photopolymerization chemistry, nanocomposite preparation, and applications in various industries are also explained to provide a comprehensive picture of the current and future capabilities of the technique and materials involved.



1. Introduction

Additive manufacturing, commonly referred to as 3D printing, is one of the most disruptive technologies of our time. AM involves "adding and joining" one layer of material on top of another to form a designed object or fabricated part.[1] AM using polymer materials (instead of ink) started in the 1980s with SLA,[2] fused deposition modeling (FDM),[3] and selective laser sintering[4] techniques. Since then, many variations in 3D printing technology have been developed such as binder jetting, digital light processing, sheet lamination, and paste extrusion, among others. A recent study[5] showed that worldwide spending on 3D printers would continuously increase in the next two years (Figure 1A). Furthermore, an estimated annual compound growth rate of 27% is expected from the \$11 billion AM industry in 2015, expanding it to \$26.7 billion by 2019.[6]

In contrast to traditional subtractive manufacturing—the process of removing materials from a bulk monolith to form a part (e.g., through cutting, grinding, machining, etc.)—AM can create more complex geometries and multicompositions. This capability changed the way people conceptualized ideas, allowing companies to save resources and product development time. Prototyping can now be accomplished within a few hours or days—a drastic improvement for a process that normally takes months using traditional methods of manufacturing. Hence, the process came to be known as "rapid prototyping." But the demand for actual parts is turning this proposition towards "rapid manufacturing." Industries are now capable of creating functional parts for end-use applications such as GE Aviation's 3D printed fuel nozzles[8] and Siemens AG's

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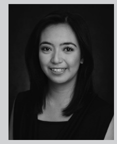
turbine blades.[9] This transformation from rapid prototyping to rapid manufacturing is, in part, possible through the development of various AM techniques and new high-performance materials. However, a recent survey[7] involving over 100 industrial manufacturers showed that there is only a very small group utilizing AM for the production of final products (Figure 1B). Moreover, a large percentage of the survey respondents still do not implement AM because it can only manufacture limited functional parts. Thus, there is a need to continuously improve existing processes and materials, or develop new ones, to widen the reach of this surprisingly underutilized technology.

Among existing AM techniques, SLA is still one of the most widely used.[10] It is based on a liquid resin photocuring process wherein liquid resin (a blend for polymerization or cross-linking) is placed in a reservoir, and a positionally programmed laser is scanned over the resin surface to initiate photopolymerization.[11] This cures the resin and converts it from liquid to solid often via a chemical crosslinking process. SLA is used in a wide variety of applications from prototyping consumer products to printing living tissues.[7] Despite these advantages, however, SLA faces two major issues: processing time and parts performance (thermo-mechanical), the latter being the focus of this paper. The layer-by-layer nature of the process prevents SLA-printed products from achieving the mechanical properties of its monolithic counterparts. One way of addressing this is through developing SLA-printed nanocomposites.

Composites are formed when two or more monolithic materials are combined such that a stronger and rigid reinforcement phase is dispersed in a weaker continuous phase, referred to as the matrix.[12] If any of the two phases have one, two, or three of its dimensions in the scale of a billionth of a meter, then the resulting material is classified as a nanocomposite.[13] SLA is headed towards direct manufacturing in the next 30 years (Figure 2) involving novel chemistry of nanocomposites or better cross-linking strategies. In the case of polymers, this would result in more thermoset or high-performance polymer properties with high thermo-mechanical strength and stability.

This review will focus on SLA as a technique to fabricate polymer nanocomposites. It starts with a detailed discussion of the process, including classifications, advantages, limitations, and essential parameters to be considered during printing. The resin chemistry and photopolymerization process involved are also reviewed to provide an understanding of changes in the material during printing. The last chapter deals with SLA nanocomposite preparation and state-of-the-art applications in bio-, structural, electronic, and magnetic materials.



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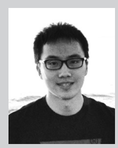

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2. Stereolithography Technique: Classifications, Advantages, Limitations, and Parameters Affecting Printing

2.1. SLA Classifications

The SLA technique may be classified according to build platform motion and laser movement.

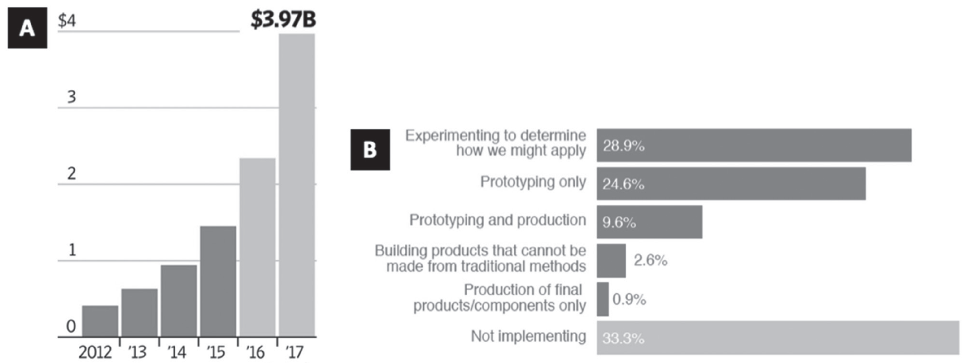

Figure 1. A) Worldwide spending on 3D printers, in billions[5] and B) 3D printing technology utilization by industrial manufacturers as of February 2014. Reproduced with permission.[7] Copyright 2014, PricewaterhouseCoopers.

2.1.1. Build-Platform Motion

There are two techniques that fall under this classification namely, top-down and bottom-up,[15,16] both of which are based on a layer-by-layer printing process (Figure 3).

In the bottom-up approach, the resin lies on top of a movable build platform. During printing, the build platform is initially placed in a position wherein only a thin layer of resin is exposed on the surface. A laser scans the exposed resin to create a cured layer with a 2D pattern. After printing one layer, the build platform moves down, and a roller provides a new layer of uncured resin. Strong interlayer adhesion is achieved by ensuring that the cure depth is greater than the resin layer thickness.

In contrast, the top-down approach utilizes a laser that scans from the bottom of the reservoir. The build platform is placed near the bottom of the reservoir such that only a thin layer of liquid resin is beneath the platform. This layer is exposed to the laser for curing, after which the build platform is lifted to allow liquid resin to refill the gap between the platform and reservoir. This is repeated layer by layer until the part is created.

The top-down approach has several advantages over a bottom-up approach. First, it is safer to use because the laser is confined within the printing device. Thus, the operator is not exposed to the laser. Second, the curing process is conducted in a sealed environment to prevent oxygen inhibition during photopolymerization. Third, recoating of liquid resin using a roller is not required because refill happens automatically with the aid of gravity. Last, a top-down approach can generate smoother printed parts due to the full contact between the liquid resin and the smooth bottom surface of the reservoir.

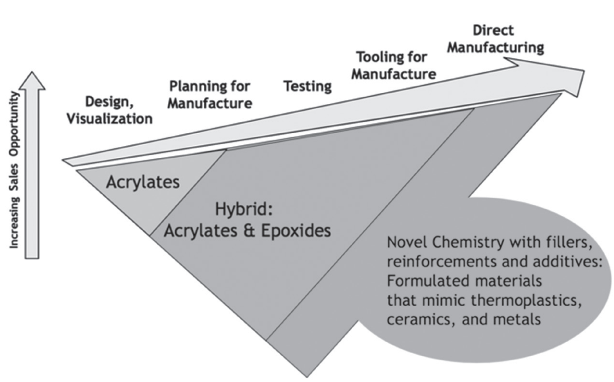

Figure 2. Market evolution of stereolithography in 30 years. Reproduced with permission.[14] Copyright 2014, RadTech.

2.1.2. Laser Movement

For the classification based on laser movement, two types of techniques are normally applied: projection-based stereolithography (PSL) and scanning-based stereolithography (SSL).[15,17,18] PSL prints every layer with a single shot of laser exposure by producing patterned laser lights and SSL scans the surface of each layer to create patterns. Thus, PSL is suitable for higher resolution printing of small parts because of the limited size of the patterned laser light. SSL, on the other hand, is good for large size printing at the cost of resolution. PSL also has a shorter printing time than SSL because each layer is printed in a single shot. Recently, researchers have developed a scanning-projection based stereolithography (SPSL) which combines both PSL and SSL. SPSL utilize a digital micromirror device (DMD), which is used to create patterned laser lights in PSL and, by moving the DMD, patterned laser lights can also scan resin surface to enable large size printing.

2.2. Advantages and Limitations

SLA affords the highest resolution among other additive manufacturing techniques. For most commercial 3D printers, the smallest features are in the range of 50–200 μm, whereas SLA printers can easily reach resolutions of 20 μm or less.[15,19] The high resolution is due to the accurate space and time control of the applied photons (intensity). Normally, one photon propagating light is

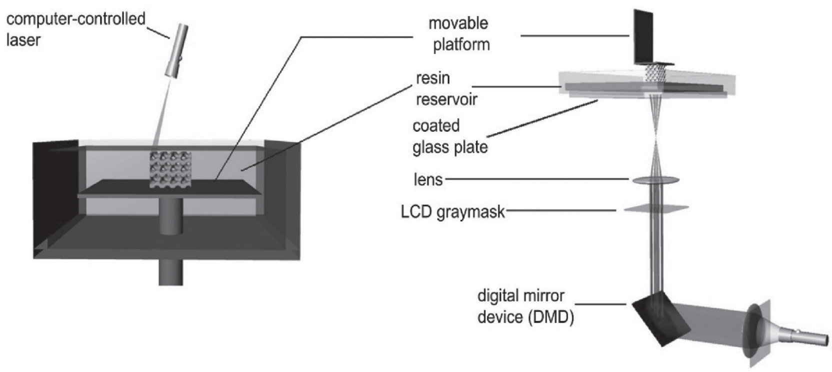

Figure 3. Schemes for bottom-up SLA (left) and top-down SLA (right). Reproduced with permission.[15] Copyright 2010, Elsevier.

applied to initiate polymerization, but two-photon[20] initiation is gaining attention because its better control can produce higher printing quality. Researchers have completed nanoscale printing quality by using two-photon laser initiation. Apart from two-photon initiation, evanescent light is also used for high-resolution printing. Evanescent light does not propagate isotropically but is localized in the near-field or even planar region which can avoid over-curing, thus enhancing printing quality.[21] Recently, Old World Laboratory[22] launched an SLA printer with a resolution up to 100 nm, which can further expand the application of SLA to print very complex and delicate parts in the sub-micrometer scale.

Despite the advantages of SLA, the technique also faces several challenges. The quality of SLA printing is limited by the "stair-stepping" effect caused by the printer's capability to fabricate only straight layers and sharp edges.[23] Some researchers[23] addressed this concern by developing slant beam rotation scanning where a rotating laser is used during printing. The inclined laser can cure resin through 360-degree angles, thus adding another degree of freedom.

Another limitation of SLA is the relatively slow printing process, which is primarily caused by low photopolymerization rates during printing. Moreover, the process is said to be inherently discontinuous.[1,24] The fundamental layer-by-layer deposition mechanism of SLA requires that laser scanning, platform moving, and resin refilling must be processed in separate, discrete (i.e., discontinuous) steps. Therefore, there is a large amount of time between each step during which no actual printing is happening. A new technique called continuous liquid interface production (CLIP) addressed this concern, allowing a part to be printed in minutes instead of hours.[25]

CLIP (Figure 4) is based on the mechanism of SLA in that it still uses laser light to initiate photopolymerization of a liquid resin, but instead of building each layer in a stepwise manner, it continuously cures the resin, thus the time between different steps is saved. The innovation in the CLIP process is the presence of a "dead zone"—a thin layer of uncured resin between the printed part and reservoir bottom. The dead zone is formed via a specially designed UV transparent and highly oxygen permeable window used as the base of the reservoir, below which pure oxygen is continuously supplied. UV transparency ensures laser penetration for resin curing, while oxygen permeability allows oxygen to penetrate the resin reservoir to inhibit polymerization. This "dead zone" is fundamental for continuous printing as it ensures that a fresh layer of resin is always present below the printed part. The thickness of the dead zone can be controlled by oxygen flux. Moreover, the build platform motion is also continuous; it moves slowly to match the curing rate of the resin. Thus, the overall printing process is conducted in a continuous manner where the laser is curing the resin while the cured resin deposits onto printed parts.

There is currently very few research involved, if any, on the use of the CLIP process for nanocomposite printing. While it is potentially a good technique for fast printing of nanocomposites with enhanced strength, several aspects still need consideration. Before photopolymerization, laser light needs to pass through unreacted resin in the dead zone. The addition of nanofillers in the unreacted resin can scatter or even block the laser, thus the intensity of the actual laser light reaching the resin may not be enough, potentially compromising print quality. Moreover, different types of resin are likely to vary the dead zone thickness with the same O_2 permeation. Thus, the dead zone thickness and O_2 penetration rate should be calibrated each time when printing with a different polymer resin.

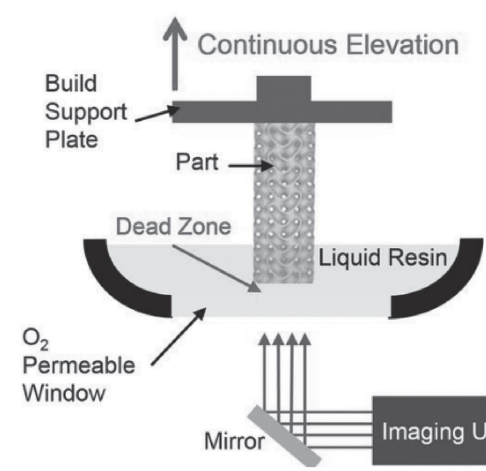

Figure 4. Schematic of continuous liquid interface production printer. Reproduced with permission.[25] Copyright 2015, AAAS

2.3. Parameters Affecting Printing

One of the applications of SLA is for printing nanocomposites. For example, complex ceramics parts may be printed by incorporating ceramic particles in a photocurable resin, followed by SLA[18] and high dielectric capacitor 3D printing with PSL.[26] However, the presence of nanoparticles introduces problems with printing accuracy because of the change in light scattering induced by the nanoparticles. It has been reported that the curing precision is related to the curing depth (C_d) and curing width (C_w), both of which determines the printing details.[27] The theoretical expression of curing depth and curing width are derived from Beer–Lambert and shown below

$$C_d = D_p \ln \frac{E}{E_C} \tag{1}$$

$$C_w = F\sqrt{\ln \frac{E}{\varphi \cdot E_C}} \tag{2}$$

where D_p is the penetration depth, E is the exposure and E_C are the critical exposure to initiate polymerization.[24] D_p is determined by the intrinsic property of resin composite, which includes the loading of nanoparticles, the size of nanoparticles and the refractive index of both nanoparticle and liquid resin. F and φ are determined by the laser beam profile and nature of resin composite (refractive index, particle size, and loading concentrations). E_C solely depends on the photoinitiator as well as liquid monomer (liquid resin). E can be determined by the equation shown below

$$E = \frac{2 \cdot p_o}{\pi \cdot w_o \cdot v_s} \tag{3}$$

where p_0 is the power of the laser at the surface of the resin, w_0 is the beam radius at e^{-2}, and v_s is the scanning speed. It is apparent from the given equations that C_d and C_w are largely influenced by the properties of nanoparticles, including loading concentration, refractive index, and size, with other parameters being a funciton of the SLA printer. For example, when the refractive index difference of nanoparticle and liquid resin is large, laser light will be significantly scattered, which will result in an insufficient D_p and the curing depth will be attenuated. The increase in light scattering will also cause more resin around the laser beam to cure. Thus, the curing width is increased, which would worsen the resolution. Therefore, work is needed to choose the appropriate nanoparticles for a specific liquid resin.[26]

The wavelength of the laser light used for SLA printing is another important parameter. Most of the laser in SLA is provided by a UV lamp, thus the wavelength range is normally between 300 and 400 nm, but this can vary depending on the SLA printer. Therefore, the photoinitiator should be chosen carefully such that it matches the wavelength of the light and absorbs it to cleave and generate radicals. Details are described in Section 3.2.

3. Stereolithography Chemistry

It is important to understand the crosslinking reaction between multifunctional prepolymers[15] in an SLA resin to modify existing resins or create new formulations tailored for specific applications. This section discusses the major components of SLA resins followed by an explanation of the stages involved in the SLA photopolymerization process.

3.1. SLA Resin Components

Several components are needed in the SLA resin to achieve high printing quality including, but not limited to, the monomer/oligomer, diluent, chain transfer agent, and photoinitiator.[1]

3.1.1. Monomer/Oligomer

Monomer/oligomers are the reactive prepolymers that undergo polymerization reaction to increase the extent of crosslinking. Most monomer/oligomers currently used for SLA are poly-acrylate derivatives where the C=C bonds in vinyl groups crosslink with different monomers. Among acrylate-based resins, methacrylate-based monomers are used widely because of their low cytotoxicity and high heat resistance.[1,28,29] Difunctional acrylates are used commonly for SLA resins, but multifunctional monomers are also applied to introduce more crosslinking sites and tune the mechanical properties of the cured resin. Moreover, the molecular weight of the monomer/oligomer largely influences the viscosity of the liquid resin. For high molecular weight resins, the viscosity is often large enough to result in a long settling time during printing, thus diluents are used to modify the viscosity. The polarity of the liquid resin is also important for nanocomposite preparation. Uniform dispersion of the nanofillers requires similar polarity with the monomer/oligomer, thus surface modification is often applied to tailor filler-resin polarity. Other types of resin include cycloaliphatic epoxies, which polymerize through a cationic polymerization process. It is worth noting that acrylate based SLA resins are prone to shrinkage after crosslinking, while epoxy based resin has better dimension stability, but has a slower curing rate.[1,29,30]

3.1.2. Diluent

Diluent agents modify the viscosity of the resin to achieve proper wetting and impregnating behavior during printing. For high quality printing, diluents must be reactive with the prepolymer and become part of the solidified resin, but must stay nonreactive under storage conditions. Adding fillers to the SLA resin can significantly increase the viscosity. Thus, diluent agents are especially important in such cases to reduce the viscosity and maintain good wetting and impregnation of resin.[1,31] For instance, 1,6 hexanediol diacrylate can be used as a diluent to sufficiently reduce the viscosity of a photopolymer resin having high solid loadings of ceramic fillers.[32]

3.1.3. Chain Transfer Agent

A chain transfer agent, specifically addition fragmentation chain transfer (AFCT) agent, is necessary for SLA resins to modify the crosslinking extent, thus regulating network formation to enhance the overall properties of printed parts. Allyl sulfides are always introduced as the efficient AFCT agents to control the molecular weight in free radical polymerization.[33] One study[34] described using allyl sulfides and methacrylate or vinyl monomers to form a crosslinking network with lower shrinkage stress and higher crosslinking density by incorporation of allyl sulfide AFCT into thiol-ene polymerization. In this case, the chain transfer process can be achieved via hydrogen abstraction in the presence of the free radicals. β-allyl sulfone was recently introduced as another well-known AFCT agent to improve the toughness of methacrylate based crosslinking networks.[35–37] Different from the thiol-ene system, the β-allyl sulfone allows free radical species to directly add to itself, followed by the fragmentation of the sulfonyl radical to continue the chain transfer process.

3.1.4. Photoinitiator

A photoinitiator is essential for photopolymerization. The prepolymer is supposed to be only reactive under a specific laser exposure, but most of the acrylate based prepolymer cannot be activated upon irradiation. This requires a proper photoinitiator that can generate reactive species under laser exposure to attack the prepolymer and initiate polymerization.[32,38] 2,2-Dimethoxy-1,2-phenylacetophenone (DMPA), because of its good solubility in monomer/oligomer and its efficient light absorbance, is a commonly used photoinitiator to absorb UV light and generate radicals.[39]

3.2. Photopolymerization Process

The SLA process involves solidification of a liquid resin into highly crosslinked networks upon light exposure. It is therefore necessary to have a proper photo initiation process and a fast crosslinking reaction for high quality printing. To understand the process better, we discuss in this section the different stages involved in the SLA photopolymerization process.

3.2.1. Photoinitiation

Free radical photoinitiators have two major types, classified according to the mechanism of free radical generation: α-cleavage and H-abstraction initiators.

The α-cleavage photoinitiator possesses the structure that will break and form two separated parts both of which contain the free radical. Such cleavage requires a specific range of UV irradiation, while different photoinitiators have different working UV lengths. As mentioned earlier, DMPA is a commonly used photoinitiator.[39] While exposed to UV irradiation, the specific bond will break and generate two free radicals as two single parts. Such photoinitiator also includes hydroxyacetophenone, benzoin, benzoin ethers, phosphineoxide, etc.[38,40–42]

On the other hand, H-abstraction photoinitiator works under a different mechanism which needs a coinitiator. When the initiator is excited by UV irradiation and steps into excited electron status, it will abstract hydrogen from the coinitiator. Thus, both the photoinitiator and the coinitiator contain free radicals. A typical representative of this type of photoinitiator is benzophenone, which has good initiation efficiency as well as good solubility in various organic solvents.[43] With different substituents on the benzene ring, benzophenone derivatives may possess better properties which help meet the needs of different applications.[43–45]

3.2.2. Excitation Transition

The excitation transition includes single photon absorption and two-photon absorption processes.[46,47] The former can only excite molecules to excited singlet state by absorbing a single photon, whereas the two-photon absorption contains three different types of sequential absorbing of two photons. The first type involves a molecule in ground state absorbing a low energy photon and reaching a higher vibrational or rotational excited state, followed by another low energy photon absorption to transfer molecules to singlet excited state. In the second type, a molecule in the ground state, after absorbing one photon to a singlet state, will further absorb another one and reach higher energy level excited states.[46,48] In the third type process, instead of directly absorbing the second photon in singlet excited state, molecules will undergo internal conversion or intersystem crossing to transfer into triplet excited state. Before decaying to the ground state, the excited molecules will then absorb the second photon and rise to higher triplet excited states T_n which are highly active.[47,49] T_n excited state is very unstable, causing the electron to soon undergo deexcitation (less than 10^{-8} s) to a lower energy orbital to release the excited energy.

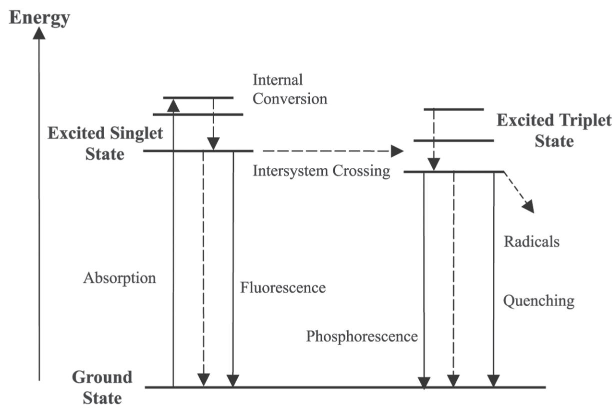

Figure 5. Jablonski Energy Diagram. Reproduced with permission.[50] Copyright 2003, Emerald Publishing Limited.

3.2.3. Deexcitation

The deexcitation process contains two photophysical processes: radiative and nonradiative. Radiative deexcitation process involves an emission of a photon from excited states, either from excited singlet state (Figure 5, fluorescence decay) or triplet state (phosphorescence), to ground state. In nonradiative deexcitation, molecules will first transfer from highly unstable singlet state then will decay to a more stable (greater than 10^{-6} s) triplet excited state by intersystem crossing or internal conversion. Because of the longer duration time in triplet excited state, a photochemical process normally happens where molecular quenches by electron transfer, bond cleavage, etc. to release excited energy as well as generate radicals or ions. Thus, the reactive species from the photoinitiator is normally formed by nonradiative deexcitation from triplet excited state.[46–50]

3.2.4. Curing Mechanism

After photoinitiation, the generated radicals will attack the functional groups in the prepolymer of the SLA resin, consequently starting solidification. The solidification process is normally via free radical polymerization,[51] which follows the steps described below

Initiation: radicals, formed from the photoinitiator, attack the prepolymer

$$\mathrm{PI(photoinitiator)} \xrightarrow{hv} I\bullet \tag{4}$$

$$\mathrm{I}\bullet + \mathrm{M} \rightarrow \mathrm{I}-\mathrm{M}\bullet \tag{5}$$

Propagation: radicals in the prepolymer attack other prepolymers for chain growth and crosslinking

$$\mathrm{I}-\mathrm{M}\bullet + \mathrm{M} \rightarrow \mathrm{I}-\mathrm{M}-\mathrm{M}\bullet \tag{6}$$

$$\mathrm{I}-\mathrm{M}-\mathrm{M}\bullet + n\mathrm{M} \rightarrow\rightarrow \mathrm{I}-\mathrm{M}_{n+1}-\mathrm{M}\bullet \tag{7}$$

Termination: Radicals in the molecules are terminated and polymerization stops.

Two stages occur during the solidification process: gelation and vitrification.[1,52] The gelation stage is normally defined as the preformation of a highly crosslinked polymer matrix of infinite molecular weight. In this stage, prepolymers partially crosslink, leading to significant increase in viscosity. Thus, liquid resin transforms into an elastic or rubber state. The gelation point is determined by the functional group, reactivity, stoichiometry, etc. of the SLA resin.[52–54] Two phases coexist during this stage: nonsoluble solid phase, composed of more crosslinked prepolymers, and a soluble liquid phase, composed of less reacted liquid resin. The unreacted monomer/oligomer can still diffuse easily within the liquid phase, thus reaction rate is predominantly kinetically controlled. As the curing process continues, more insoluble liquid phase is being generated while the soluble liquid phase decreases.[52–56] When viscosity and crosslinking density increase to a certain point, vitrification stage occurs where SLA resin transforms from rubbery state to glassy state. The diffusion of free monomer/oligomer becomes difficult as the amount of soluble liquid phase is very limited. Therefore, the reaction rate largely decreases and becomes diffusion-controlled rather than kinetically controlled. This diffusion-controlled reaction also determines the final conversion degree, and influence the homogeneity of the final cured matrix.[57,58] Brittleness is often observed in SLA-printed structures, which can be due to the inhomogeneous crosslinked network, resulting from the un-even diffusion of unreacted monomer/oligomers in the vitrification stage or the fast reaction rate in the gelation stage.[59,60]

4. SLA Polymer Nanocomposites

4.1. Nanocomposite Preparation

Different methods have been developed to prepare nanocomposites for SLA. These methods vary depending on the nature of the nanofiller and the resin. The primary goal is to achieve a homogeneous mixture with an acceptable viscosity to obtain high quality printed products. Homogeneity may be achieved by mechanical mixing, shear mixing, sonication, ultrasonication, or a combination of these methods. Sonication may be sufficient for less viscous mixtures or those having relatively low solids loading.[61] On the other hand, ultrasonication is more applicable for mixtures with high viscosity or high solids loading. Some researchers working on epoxy-based resin filled with multiwalled carbon nanotubes[62] and nanosilica-reinforced resin[63] used a combination of mechanical mixing and ultrasonication to gradually disperse the nanofiller in the SLA resin and prevent agglomeration. Agglomeration is expected with such nanoparticles due to their high tendency for self-association.[12] A homogenizer[64] or high shear mixer[63] may also be used, with the latter often coupled with ultrasonication to achieve a homogeneous mixture.

Homogeneity can also be achieved by controlling resin viscosity. One study[65] used a viscosity-increasing agent to prevent particle agglomeration by facilitating increased viscous drag and plastic fluidity. This resulted in a stable dispersion, lasting for more than 10 d.[65] However, the viscosity must not be increased to the point that flow becomes difficult as this will pose challenges during printing when recoating with a fresh layer of the resin after each layer. A flat resin surface is also difficult to achieve with highly viscous materials because gravity will no longer be capable of producing a flat surface.[66]

There are instances when the filler material has poor affinity with the resin, thus producing a heterogeneous mixture even after performing the mixing methods explained above. This may be addressed through surface modification of the nanomaterial. A study[67] working on nanosilica/acrylic nanocomposites was able to organically modify the nanosilica surface to improve the stability of the resulting resin dispersion. The modification involved methacrylate groups which allowed covalent bonding between inorganic particles and the polymeric matrix. Similarly, organically modified montmorillonite has been seen to improve the mobility of propagating chains to increase exothermic reaction and cure rate of the systems.[68]

Once a homogeneous and stable resin mixture is successfully used for printing, it is usually necessary for the printed part to undergo postprocessing. Postprocessing prepares the printed part for its intended application.[1] Common postprocessing techniques for SLA include rinsing the part in isopropyl alcohol (IPA) to remove uncured resin, support removal, and UV or thermal postcuring to facilitate further curing of the polymer. Other postprocessing methods impart additional functionality to the printed part, such as magnetic property, as will be discussed in the following section.

4.2. Nanocomposite Applications

SLA has a wide range of applications once a homogeneous resin is achieved and a successful print obtained. Here we present an overview of current advances in SLA polymer nanocomposite applications, specifically in the fields of bio-, structural, and electronic and magnetic materials.

4.2.1. Biomaterials

The increasing need for patient-specific medical devices has driven 3D printing forward in the healthcare market. Global revenues as of 2014 reached $487 million and are expected to grow at a rate of 18.3% until 2020.[69] From biomodels,[70] organs-on-chip,[71] to healthcare centers fully equipped with 3D printing facilities,[72] the technology is changing the way medical practitioners approach healthcare challenges. Materials commonly utilized in this area include metals and ceramics in the dental and prosthetics industries, while polymers are prevalent in scaffolds and organ manufacture, among others. As of 2013, 35% of the overall market share was accounted to polymers with photopolymerization technology being one of the most utilized techniques.[73]

Well-known polymers used for 3D printing biomaterials include polyethylene glycol diacrylate (PEGDA), polylactic acid (PLA), and polycaprolactone (PCL). PEGDA, a commonly used material for hydrogels, has been studied widely for scaffolding applications. Among PEGDA's attractive properties are its ability to retain 3D geometry despite hydration[74] and its inherent photosensitivity.[75] Unfortunately, PEGDA has low biological activity and its intrinsic hydrophobicity makes cell adhesion an issue, consequently leading to poor tissue formation.[74] To enhance biocompatibility, some researchers[76] have successfully incorporated nanocrystalline hydroxyapatite (nHA)— the primary component in bone— as filler. In one study,[76] an nHA-PEGDA bio-ink was developed with arginine-glycine-aspartic acid-serene (RGDS) peptide as cell adhesive. The nHA used was ≈25 nm wide and 50–100 nm long and was obtained via wet chemistry method and hydrothermal treatment. The resulting photosensitive bio-ink was then printed using a custom-made table-top SLA bioprinter. Moreover, low-intensity pulsed ultrasound (LIPUS) was used to promote cell regeneration further. The study found that the combined effect of the bioactive 3D printed scaffolds and LIPUS process improved "cell proliferation, alkaline phosphatase activity, calcium deposition, and total protein content."[76] Furthermore, Young's modulus increased by as much as 150% for the nHA-PEGDA and nHA-PEGDA-RGDS nanocomposites.[76] This shows the capability of nHA fillers to not only enhance the biocompatibility of PEGDA, but also to enhance mechanical properties.

Unlike PEGDA, PLA and PCL have high biological activity. A major drawback in using these polymers, however, is the fact that they are not photosensitive. Functionalization with methacrylate-, diacrylate-, or fumarate- groups is required to obtain a photopolymer resin.[77] Some researchers use indirect SLA instead where a sacrificial mold is fabricated using conventional SLA and then the biocompatible polymer of choice is injected into the mold and allowed to cure. One study[78] used this technique to fabricate poly-L-lactide acid and PCL scaffolds and found that the resulting parts had good resolution. It was also observed through GPC that the indirect SLA process did not affect the molecular weight of the biomaterial. Biocompatibility of the scaffolds was also studied through cytotoxicity tests. Results proved to be promising with cell viability reaching more than 90% within a week.[78] Fillers typically used for PCL and PLA nanocomposites include alginates,[79] hydroxyapatite,[80] and chitosan.[81] Commercially available alginates may be used such as the low-viscosity and high G-content nonmedical grade LF10/60 alginate, with $CaCl_2$ as a crosslinker.[79]

Researchers are now exploring bio-based SLA resins as an alternative to petroleum-based polymers discussed above. One study[77] used soybean oil epoxidized acrylate to create scaffolds possessing shape-memory effect. Cytotoxicity results showed improved cell adhesion and proliferation compared with PEGDA, while the scaffolds had similar performance with those made from PCL and PLA. An advantage of using naturally derived resins, such as those from plant oil, is that they are both biocompatible and the presence of reactive unsaturated double bonds makes them easier to functionalize than PCL and PLA.[77] No filler was used in the said study, but given that its performance is comparable to PCL and PLA, soybean oil epoxidized acrylate becomes a promising material to be reinforced by typical fillers compatible with PCL and PLA such as hydroxyapatite, chitosan, and similar nanomaterials.

Overall, 3D printing in the healthcare industry has come a long way. Challenges are still present, especially those linked with having 3D printed parts mimic actual biological environments.[76] Fortunately, innovations in materials selection, 3D printer equipment design, and postprocessing procedures offer promising opportunities for this technology.

4.2.2. Structural Materials

3D printed structural components have gained popularity in various industries including, but not limited to, aerospace,[84] automotive,[85] and microcomponents for electronic devices.[86] Although the material used in this area are predominantly metals because of their superior mechanical properties, research on tailoring polymer nanocomposites for structural applications are also present. For instance, a recent study[87] successfully demonstrated a 3D printed component having a negative thermal expansion fabricated using multimaterial projection microstereolithography. The composite lattice was composed of PEGDA and copper nanoparticle-reinforced beams. The average size of copper nanoparticles used was 50–80 nm. It was observed that the coefficient of thermal expansion (CTE) of PEGDA decreases with increasing copper nanoparticle concentration. This was attributed to copper's low CTE and high bulk modulus compared to PEGDA.[87] Consequently, the 3D composite lattices were designed in such a way that the PEGDA beams would expand faster, causing the reinforced PEGDA beams to bend inward, resulting in a smaller overall volume upon heating. This material can be used for applications where thermal stress is an issue such as in microcomponents of electronic devices that must have little to no expansion even as the device heats up with continuous use.

Aside from 3D printed metal–polymer nanocomposites, ceramic components have been printed using polymer-derived ceramics (PDCs)—a suspension of ceramic particles dispersed in a polymer matrix and cured using UV light. Studies on photopolymerization of PDCs have existed since the 1980s, but it is only recently that people started considering it for AM. This material has potential applications in creating strong, complex-shaped objects capable of withstanding high temperature and harsh environments such as for "micro-electromechanical systems (MEMS), device packaging, propulsion, and thermal protection systems".[91] There are three general stages in the 3D printing of ceramic parts from polymers: (1) powder and body formulation, (2) production of green (i.e., unsintered) shape via AM processes, and (3) sintering, microstructure development, and finishing.[88] Focusing on the second stage, there are several methods to print the green shape including conventional SLA (also referred to as lithography-based ceramic manufacturing or LCM) and self-propagating photopolymer waveguide (SPPW) technology. SPPW involves a carefully chosen monomer that alters the index of refraction of the material after photopolymerization, causing UV light internal reflection, allowing continuous curing of layers.[91] This "UV trapping" effect makes SPPW a much faster process than conventional SLA. Both methods have been successful in fabricating polymer-derived stereolithographic ceramic components. For instance, one study[90] used a commercially available methyl-silsesquioxane resin to prepare the preceramic photopolymer. The said resin was mixed with solvent free silicone powder having a $(CH_3-SiO_{3/2})_x$ structure. The powder was dissolved in the resin by stirring and sonication in a solvent mixture of tetrahydrofuran (THF) and tripropylene glycol monomethyl ether.[90] Cellular geometries (Figure 6C) having strut sizes around 200 µm were successfully fabricated from the preceramic photopolymer via LCM technique. Subsequent pyrolysis yielded SiOC parts with superior surface quality.[90] On the other hand, a separate study[91] used (mercaptopropyl) methylsiloxane with vinylmethoxysiloxane, a photoinitiator, and an inhibitor in formulating a photosensitive siloxane resin for SPPW. Using a patterned mask, the resulting resin was exposed to UV light to form large microlattice and honeycomb structures. Pyrolysis resulted to high-density structures with excellent surface finish and superior compressive strength compared to commercial SiC foams of the same density.[91]

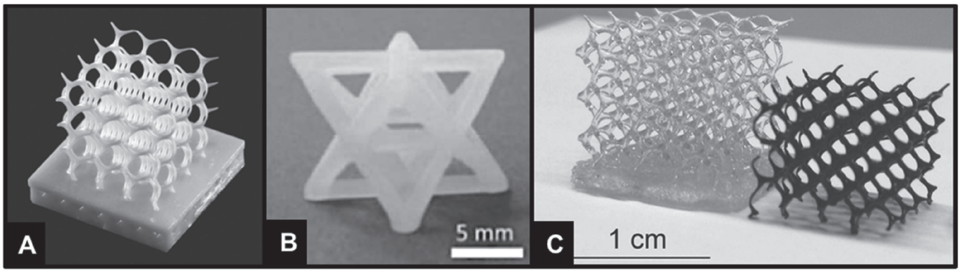

Figure 6. A) Lattice structure in an Al_2O_3-fixed digital mirror device. Reproduced with permission.[88] Copyright 2016, Annual Reviews. B) 3D printed octet truss. Reproduced with permission.[89] Copyright 2016, American Chemical Society. C) A green body structure from additive photopolymerization and the corresponding pyrolyzed 3D ceramic component. Reproduced with permission.[90] Copyright 2016.

4.2.3. Electronic and Magnetic Materials

The area of 3D printed electronic and magnetic materials is an emerging field with applications in sensors, actuators, MEMS, and nanocomposite scaffolds. Researchers are constantly finding and improving ways to 3D print parts that are not only structurally sound, but functionalized with specific materials to impart conductivity, magnetic properties, or both. The filler often used to achieve magnetic 3D printed parts is magnetite nanoparticles (MNPs) because of its chemical stability and biocompatibility.[92,93] These parts may be prepared by either utilizing a postprocessing technique upon printing or by direct incorporation of conductive or magnetic fillers into the resin to achieve a functionalized SLA resin.[94] For magnetic parts, a common postprocessing technique used is electroless plating, which involves depositing a thin layer of magnetic alloy on an SLA-printed part. A disadvantage of this technique is the requirement of an additional step after printing to functionalize the material.[94] Hence, the second method of filler incorporation is more convenient since it involves only a single step.

As an example, ferromagnetically responsive cantilever-type composite microstructures were fabricated successfully via SLA in a recent study.[94] Magnetic iron (II, III) oxide nanopowder was used with diameters ranging from 50 to 100 nm. The photopolymer resin formulation consisted of a bifunctional monomer SR349 with Lucirin-TPO (trimethylbenzoyl diphenylphosphine oxide) as a photoinitiator. Iron oxide was dispersed in the resin via mechanical stirring for 4 h to achieve homogeneity. Qualitative evaluation of magnetic sensitivity showed good sensing performance, making it a promising material for sensors and actuators.[94]

MNP-reinforced impellers for flow sensors (Figure 7) have also been printed successfully via microstereolithography.[95] Commercially available magnetite nanoparticles having an average diameter of 50 nm was used together with a photopolymer resin consisting of an acrylic oligomer, crosslinking agents, a photoinitiator, and a dye to act as an inhibitor. The monomer to crosslinker ratio was controlled such that the resulting resin would be viscous enough to prevent nanoparticle agglomeration.[95] The resulting impeller proved to be functional after evaluating its rotation using an external magnetic field sensor.

In relation to biomaterials, the medical industry is now looking at magnetic nanocomposite scaffolds, which have the benefit of triggering tissue growth using magnetic fields to guide bone regeneration.[96] One study[83]

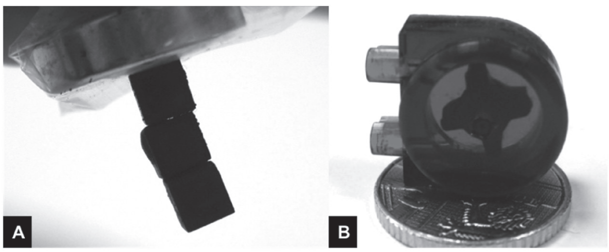

Figure 7. A) 3D printed cubes suspended from a neodymium magnet and B) 3D printed flow sensor. Reproduced with permission.[95] Copyright 2011, Elsevier.

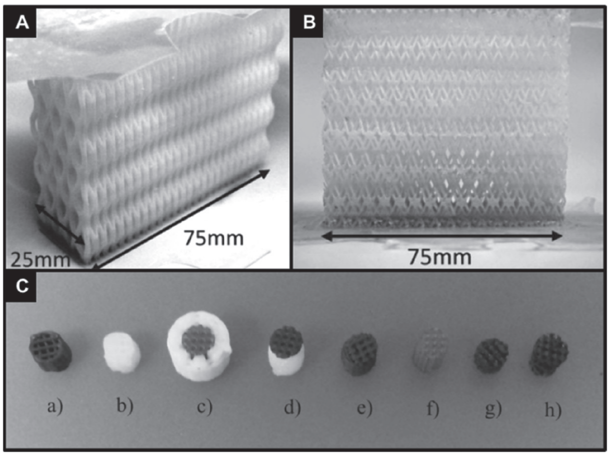

Figure 8. A) A single hollow lattice with sacrificial scaffold. Reproduced with permission.[82] B) Two interlaced lattices, with sacrificial scaffold formed via photopolymer waveguide process. Reproduced with permission.[82] C) Scaffolds printed using FDM, SLA, and a combination of both. Reproduced with permission.[83] Copyright 2015, Springer.

used a combination of FDM and SLA techniques to 3D print magnetic coaxial and bilayer scaffold structures (Figure 8C). The thinner scaffold layer for cartilage regeneration was printed using SLA, while the bone regeneration layer was printed via FDM using a highly reinforced polymer matrix. A PEGDA/MNP composite was printed for the SLA layer of the scaffold. The MNP powder used had a particle size of about 200 nm, while the photopolymer resin was a mixture of PEGDA and Lucirin-TPO as a photoinitiator. Mixing was carried out via sonication to ensure a homogeneous dispersion of the nanofiller in the resin.[83] Aside from imparting magnetic properties to the polymer resin, the addition of MNPs was shown to improve the mechanical properties of the scaffold as evidenced by the increase in compressive modulus of the nanoparticle-reinforced scaffolds. Furthermore, the study is a good example where the high resolution of SLA becomes essential for printing the detailed structure for the thin layer of the scaffold.

Regarding conductivity, graphene-based materials have been widely studied as filler to functionalize SLA resins. Graphene is a well-known conductor, but is incapable of interacting well with polymers because of its surface property. Hence, the usual route taken by researchers is to use graphene oxide (GO) as starting material and then reducing GO to graphene, through thermal or chemical means during or after printing, to recover conductivity. The oxygenated functional groups present in GO allow better interaction with most polymer resins, thus achieving a homogenous and stable mixture—vital for successful SLA printing. A recent study[97] reinforced a commercially available photopolymer resin with GO (prepared using modified Hummers' method) and successfully printed a scaffold structure using projection stereolithography (PSLA). The electrical conductivity of the printed part was achieved by pyrolysis to reduce GO to graphene. Resulting conductivities were found to be comparable to those of semiconductors, but at the expense of mechanical properties due to thermal reduction. An alternative GO reduction method might be in-situ reduction using hydrazine,[98] which also eliminates any post-processing steps.

Summary

Stereolithography was presented in this review as a technique for fabricating polymer nanocomposites. Variations of the technique were first discussed, followed by its advantages and disadvantages. Innovations addressing major drawbacks of the technology were also highlighted, including the CLIP technology. A comprehensive discussion involving photopolymerization chemistry during printing was also presented. Preparation of SLA polymer nanocomposites was then discussed with emphasis on techniques to achieve a homogeneous resin mixture before and during printing. In the last section, current developments in SLA polymer nanocomposite applications were presented, showing how using different nanofillers can potentially yield functional materials for biomedical, structural, electronic and magnetic applications, among others.

Stereolithography will have an increasingly wider adaptation as a manufacturing process if more nanocomposite materials can be incorporated in the technology. There are promising new materials, such as plant-based resins, where nanofillers may potentially be incorporated to impart improved thermo-mechanical performance and other field-responsive or even stimuli-responsive properties. Integration of SLA with other additive manufacturing techniques, such as FDM, also seems promising as this will enable the efficient fabrication of more complex designs capable of producing products with properties not attainable from a single technique alone.

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